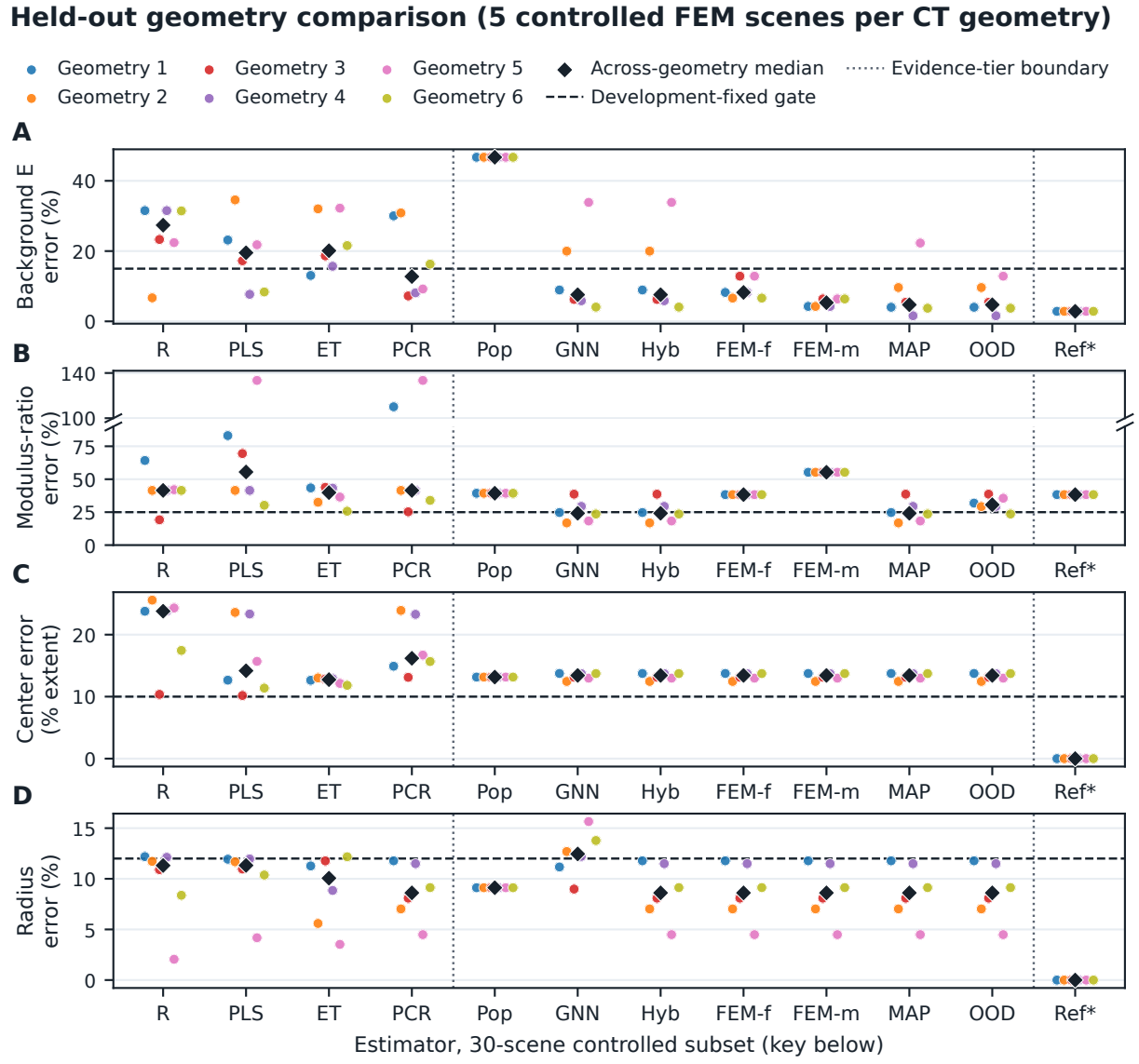

Estimator key

R = Ridge
PLS = Partial least squares
ET = ExtraTrees
PCR = PCA–Ridge
Pop = Training population
GNN = Raw MeshGNN
Hyb = Uncalibrated hybrid initializer
FEM-f = Fixed-start FEM
FEM-m = Deterministic multistart FEM
MAP = Uncalibrated hybrid-prior screened MAP
OOD = Uncalibrated hybrid-prior OOD fallback
Ref* = True-region/true-force reference